Packet Tracer - Propagate a Default Route in OSPFv2

Addressing Table

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
R1	G0/0	172.16.1.1	255.255.255.0	N/A
	S0/0/0	172.16.3.1	255.255.255.252	
	S0/0/1	192.168.10.5	255.255.255.252	
R2	G0/0	172.16.2.1	255.255.255.0	N/A
	S0/0/0	172.16.3.2	255.255.255.252	
	S0/0/1	192.168.10.9	255.255.255.252	
	S0/1/0	209.165.200.225	255.255.255.224	
R3	G0/0	192.168.1.1	255.255.255.0	N/A
	S0/0/0	192.168.10.6	255.255.255.252	
	S0/0/1	192.168.10.10	255.255.255.252	
PC1	NIC	172.16.1.2	255.255.255.0	172.16.1.1
PC2	NIC	172.16.2.2	255.255.255.0	172.16.2.1
PC3	NIC	192.168.1.2	255.255.255.0	192.168.1.1
Web Server	NIC	64.100.1.2	255.255.255.0	64.100.1.1

Objectives

Part 1: Propagate a Default Route

Part 2: Verify Connectivity

Background

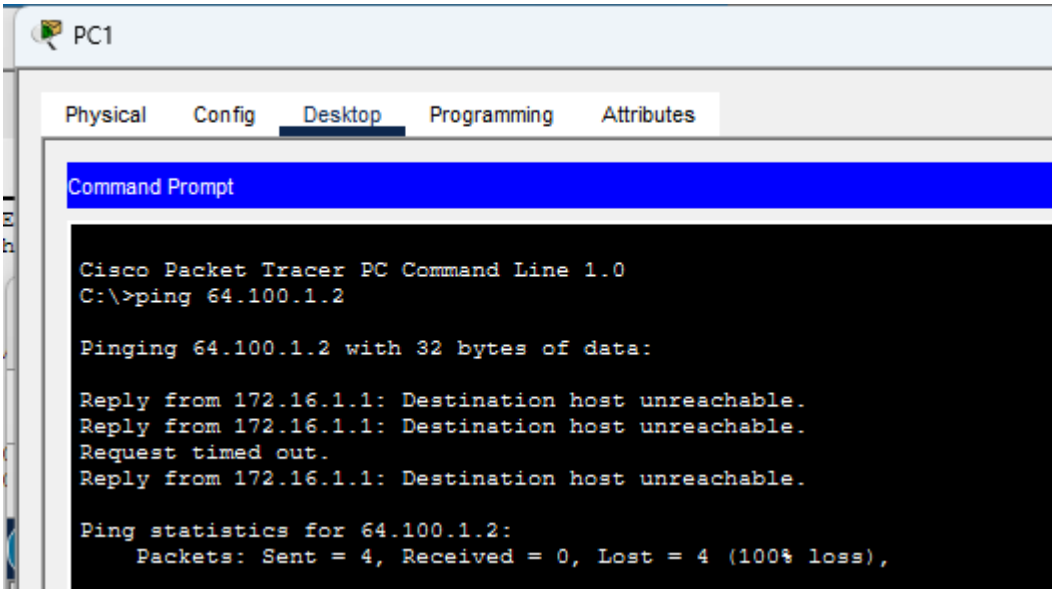
ในกิจกรรมนี้ คุณจะกำหนด IPv4 default route ไปยังอินเทอร์เน็ต และเผยแพร่ default route นั้นไปยัง routers อื่น ๆ ที่ใช้ OSPF จากนั้นคุณจะตรวจสอบว่า default route อยู่ในตารางการกำหนดเส้นทางของ routers ท้ายน้ำ และโฮสต์สามารถเข้าถึง web server บนอินเทอร์เน็ตได้

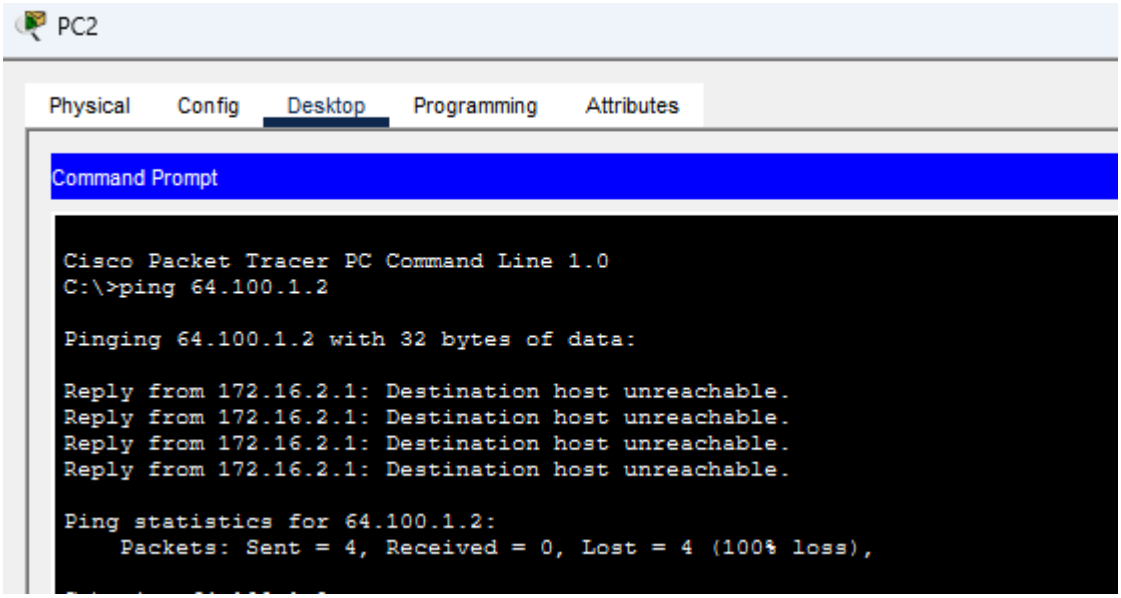
Instructions

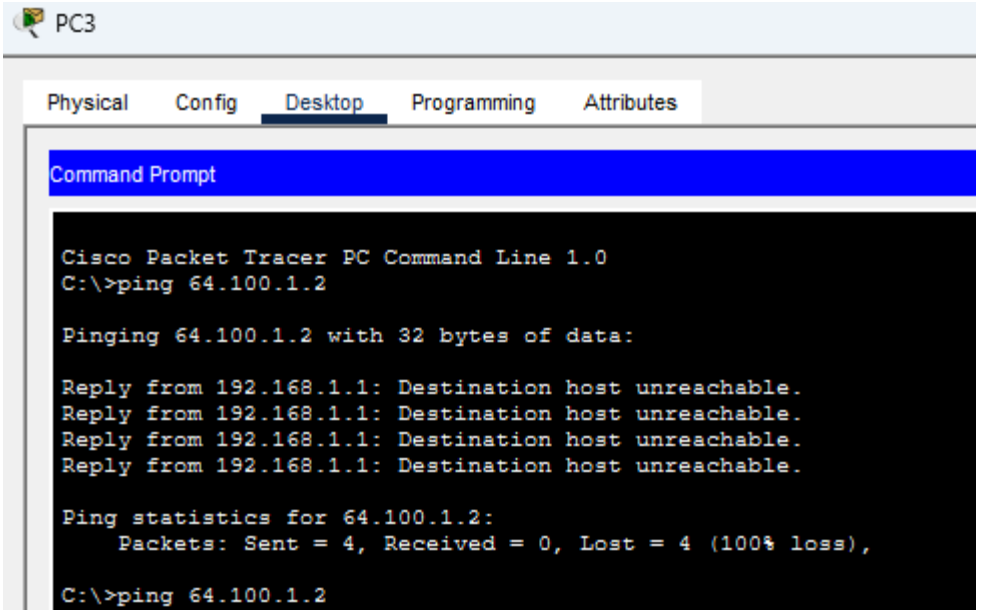
Part 1: Propagate a Default Route

Step 1: Test connectivity to the Web Server

a. จาก PC1 PC2 และ PC3 ให้พยายาม ping ไปยัง Web Server ที่ IP address 64.100.1.2







Were any of the pings successful?

What message did you receive, and which device issued the message?

b. ตรวจสอบตารางการกำหนดเส้นทาง (routing table) บน routers R1 R2 และ R3

```
R1>
R1>enable
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
C       172.16.1.0/24 is directly connected, GigabitEthernet0/0
L       172.16.1.1/32 is directly connected, GigabitEthernet0/0
O       172.16.2.0/24 [110/65] via 172.16.3.2, 00:06:43, Serial0/0/0
C       172.16.3.0/30 is directly connected, Serial0/0/0
L       172.16.3.1/32 is directly connected, Serial0/0/0
O    192.168.1.0/24 [110/65] via 192.168.10.6, 00:06:43, Serial0/0/1
     192.168.10.0/24 is variably subnetted, 3 subnets, 2 masks
C       192.168.10.4/30 is directly connected, Serial0/0/1
L       192.168.10.5/32 is directly connected, Serial0/0/1
O       192.168.10.8/30 [110/128] via 192.168.10.6, 00:06:43, Serial0/0/1
                        [110/128] via 172.16.3.2, 00:06:43, Serial0/0/0
     209.165.200.0/27 is subnetted, 1 subnets
O       209.165.200.224/27 [110/128] via 172.16.3.2, 00:06:43, Serial0/0/0
```

```
R2>enable
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
O       172.16.1.0/24 [110/65] via 172.16.3.1, 00:07:26, Serial0/0/0
C       172.16.2.0/24 is directly connected, GigabitEthernet0/0
L       172.16.2.1/32 is directly connected, GigabitEthernet0/0
C       172.16.3.0/30 is directly connected, Serial0/0/0
L       172.16.3.2/32 is directly connected, Serial0/0/0
O    192.168.1.0/24 [110/65] via 192.168.10.10, 00:07:26, Serial0/0/1
     192.168.10.0/24 is variably subnetted, 3 subnets, 2 masks
O       192.168.10.4/30 [110/128] via 172.16.3.1, 00:07:26, Serial0/0/0
                        [110/128] via 192.168.10.10, 00:07:26, Serial0/0/1
C       192.168.10.8/30 is directly connected, Serial0/0/1
L       192.168.10.9/32 is directly connected, Serial0/0/1
     209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C       209.165.200.224/27 is directly connected, Serial0/1/0
L       209.165.200.225/32 is directly connected, Serial0/1/0

R2#configure terminal
```

```
R3#
R3#R2>enable
Translating "R2>enable"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.10.9 to network 0.0.0.0

     172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
O       172.16.1.0/24 [110/65] via 192.168.10.5, 00:19:17, Serial0/0/0
O       172.16.2.0/24 [110/65] via 192.168.10.9, 00:19:17, Serial0/0/1
O       172.16.3.0/30 [110/128] via 192.168.10.9, 00:19:17, Serial0/0/1
                      [110/128] via 192.168.10.5, 00:19:17, Serial0/0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
     192.168.10.0/24 is variably subnetted, 4 subnets, 2 masks
C       192.168.10.4/30 is directly connected, Serial0/0/0
L       192.168.10.6/32 is directly connected, Serial0/0/0
C       192.168.10.8/30 is directly connected, Serial0/0/1
L       192.168.10.10/32 is directly connected, Serial0/0/1
     209.165.200.0/27 is subnetted, 1 subnets
O       209.165.200.224/27 [110/128] via 192.168.10.9, 00:19:17, Serial0/0/1
```

What statement is present in the routing tables that indicates that the pings to the Web Server will fail?

Step 2: Configure a default route on R2.

Configure **R2** with a directly attached default route to the Internet.

R2(config)# **ip route 0.0.0.0 0.0.0.0 Serial0/1/0**

```
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip route 0.0.0.0 0.0.0.0 Serial0/1/0
%Default route without gateway, if not a point-to-point interface, may impact performance
```

Note: Router จะแสดงคำเตือนว่า หากอินเทอร์เฟซนี้ไม่ใช่ point-to-point connection อาจส่งผลต่อประสิทธิภาพ คุณสามารถละเว้นคำเตือนนี้ได้เพราะเป็น point-to-point connection

Step 3: Propagate the route in OSPF.

กำหนด OSPF ให้เผยแพร่ default route ในการอัปเดตการกำหนดเส้นทาง OSPF

```
R2(config)# router ospf 1
R2(config-router)# default-information originate
```

```
R2(config)#router ospf 1
R2(config-router)#default-information originate
R2(config-router)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R2#
```

Step 4: Examine the routing tables on R1 and R3.

Examine the routing tables of **R1** and **R3** to verify that the route has been propagated.

```
R1> show ip route
<output omitted>
Gateway of last resort is 172.16.3.2 to network 0.0.0.0
<output omitted>
O*E2 0.0.0.0/0 [110/1] via 172.16.3.2, 00:00:08, Serial0/0/0
!-------------------
R3> show ip route
<output omitted>
Gateway of last resort is 192.168.10.9 to network 0.0.0.0
<output omitted>
O*E2 0.0.0.0/0 [110/1] via 192.168.10.9, 00:08:15, Serial0/0/1
```

```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.10.9 to network 0.0.0.0

     172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
O       172.16.1.0/24 [110/65] via 192.168.10.5, 00:19:17, Serial0/0/0
O       172.16.2.0/24 [110/65] via 192.168.10.9, 00:19:17, Serial0/0/1
O       172.16.3.0/30 [110/128] via 192.168.10.9, 00:19:17, Serial0/0/1
                      [110/128] via 192.168.10.5, 00:19:17, Serial0/0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
     192.168.10.0/24 is variably subnetted, 4 subnets, 2 masks
C       192.168.10.4/30 is directly connected, Serial0/0/0
L       192.168.10.6/32 is directly connected, Serial0/0/0
C       192.168.10.8/30 is directly connected, Serial0/0/1
L       192.168.10.10/32 is directly connected, Serial0/0/1
     209.165.200.0/27 is subnetted, 1 subnets
O       209.165.200.224/27 [110/128] via 192.168.10.9, 00:19:17, Serial0/0/1
 --More--
```

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

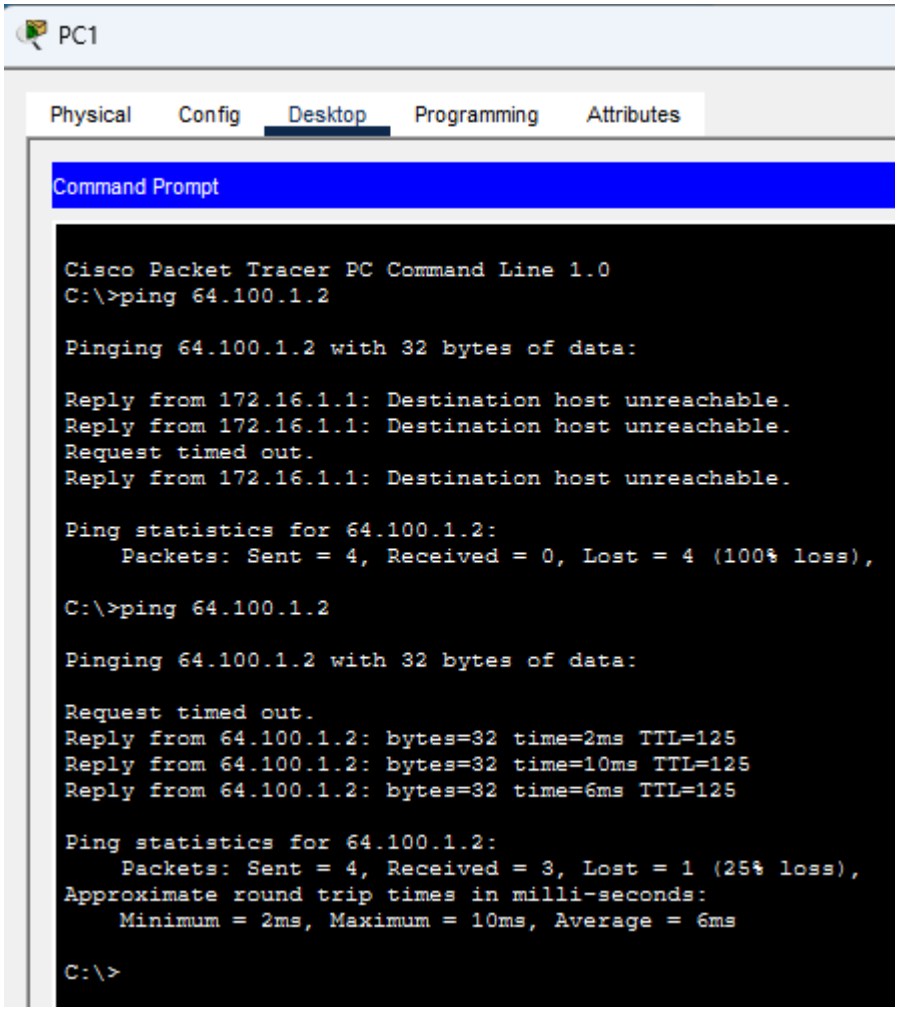
Gateway of last resort is 172.16.3.2 to network 0.0.0.0

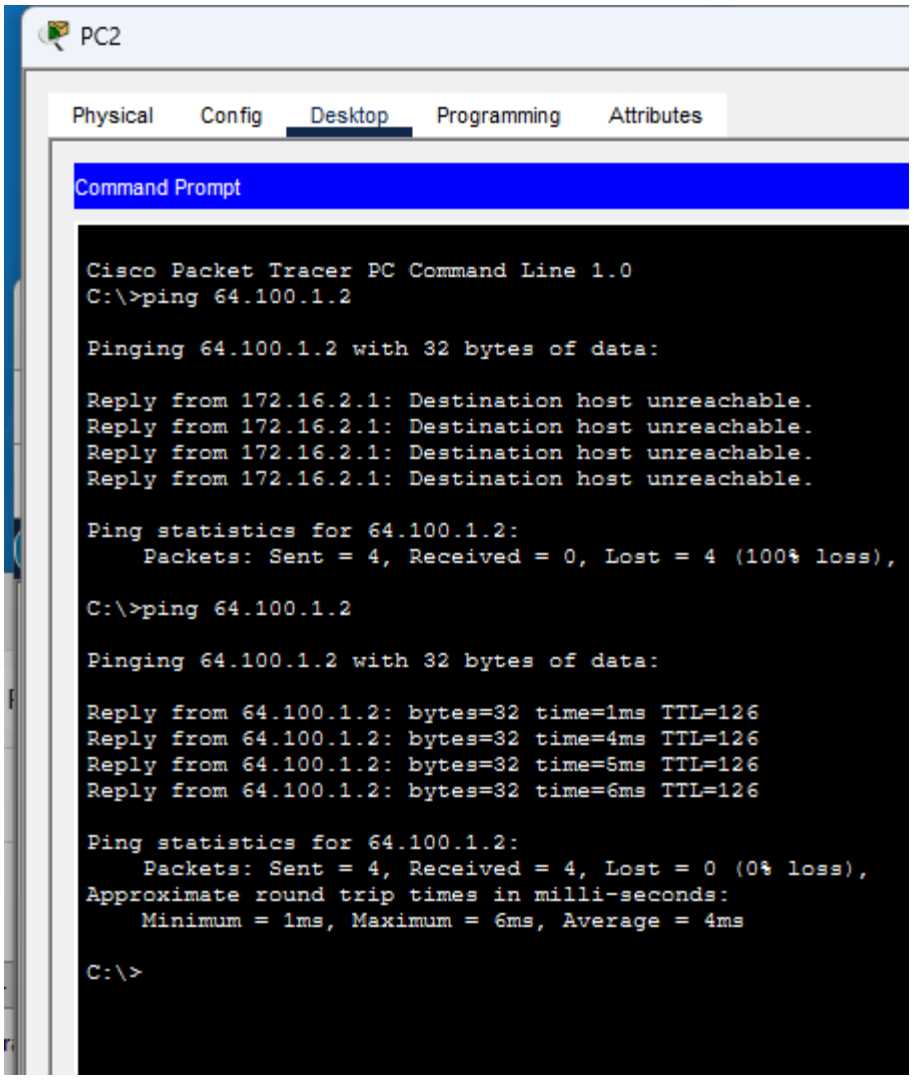
     172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
C       172.16.1.0/24 is directly connected, GigabitEthernet0/0
L       172.16.1.1/32 is directly connected, GigabitEthernet0/0
O       172.16.2.0/24 [110/65] via 172.16.3.2, 00:11:01, Serial0/0/0
C       172.16.3.0/30 is directly connected, Serial0/0/0
L       172.16.3.1/32 is directly connected, Serial0/0/0
O    192.168.1.0/24 [110/65] via 192.168.10.6, 00:11:01, Serial0/0/1
     192.168.10.0/24 is variably subnetted, 3 subnets, 2 masks
C       192.168.10.4/30 is directly connected, Serial0/0/1
L       192.168.10.5/32 is directly connected, Serial0/0/1
O       192.168.10.8/30 [110/128] via 192.168.10.6, 00:11:01, Serial0/0/1
                        [110/128] via 172.16.3.2, 00:11:01, Serial0/0/0
     209.165.200.0/27 is subnetted, 1 subnets
O       209.165.200.224/27 [110/128] via 172.16.3.2, 00:11:01, Serial0/0/0
O*E2 0.0.0.0/0 [110/1] via 172.16.3.2, 00:01:51, Serial0/0/0

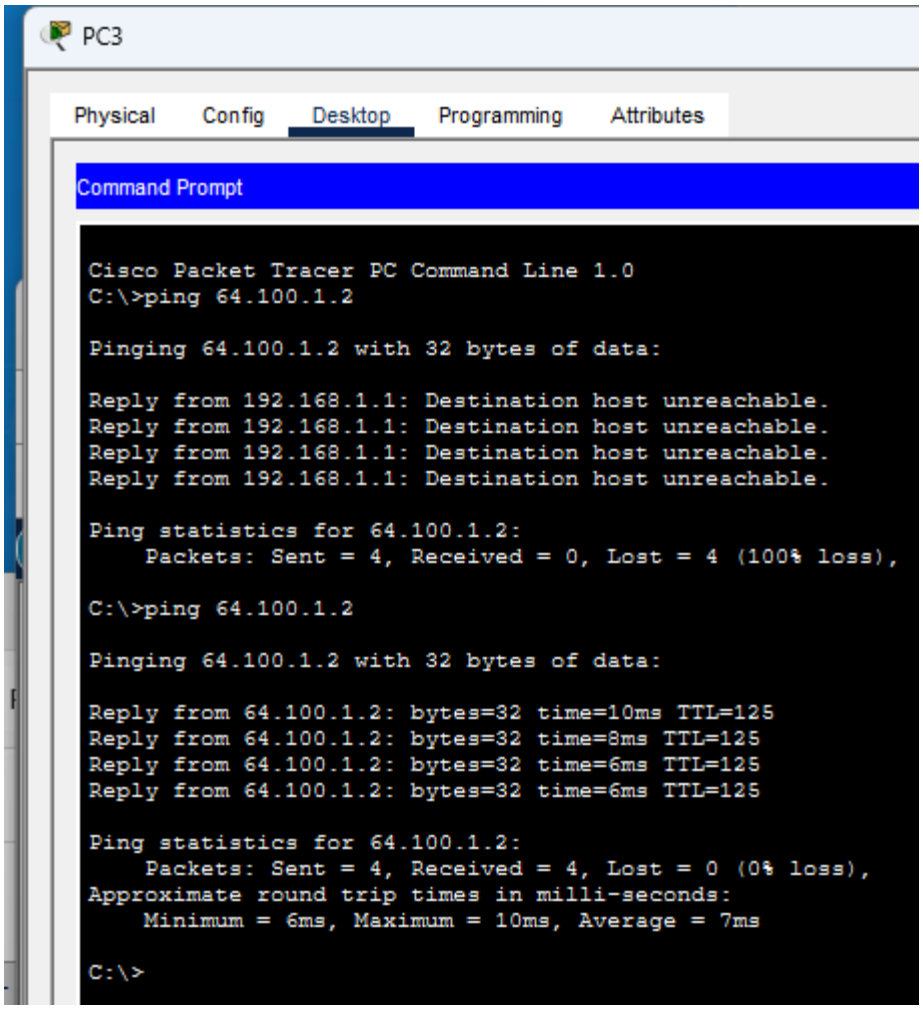
R1#
```

Part 2: Verify Connectivity

Verify that **PC1**, **PC2**, and **PC3** can ping the web server.




PC2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 64.100.1.2
Pinging 64.100.1.2 with 32 bytes of data:
Reply from 172.16.2.1: Destination host unreachable.
Reply from 172.16.2.1: Destination host unreachable.
Reply from 172.16.2.1: Destination host unreachable.
Reply from 172.16.2.1: Destination host unreachable.
Ping statistics for 64.100.1.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 64.100.1.2
Pinging 64.100.1.2 with 32 bytes of data:
Reply from 64.100.1.2: bytes=32 time=1ms TTL=126
Reply from 64.100.1.2: bytes=32 time=4ms TTL=126
Reply from 64.100.1.2: bytes=32 time=5ms TTL=126
Reply from 64.100.1.2: bytes=32 time=6ms TTL=126
Ping statistics for 64.100.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 6ms, Average = 4ms
C:\>


PC3
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 64.100.1.2
Pinging 64.100.1.2 with 32 bytes of data:
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Ping statistics for 64.100.1.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 64.100.1.2
Pinging 64.100.1.2 with 32 bytes of data:
Reply from 64.100.1.2: bytes=32 time=10ms TTL=125
Reply from 64.100.1.2: bytes=32 time=8ms TTL=125
Reply from 64.100.1.2: bytes=32 time=6ms TTL=125
Reply from 64.100.1.2: bytes=32 time=6ms TTL=125
Ping statistics for 64.100.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 6ms, Maximum = 10ms, Average = 7ms
C:\>